Contact

sandy91@gmail.com

www.linkedin.com/in/sandeepsingh91 (LinkedIn)

Top Skills

Statistics
Statistical Modeling
Technical Leadership

Languages

English
Hindi

Certifications

AWS Certified Machine Learning – Specialty
AI Engineer Agentic Track: The Complete Agent & MCP Course
Object Oriented Programming Using C#
Getting and Cleaning Data
Team Leadership

Honors-Awards

Certificate of Recognition | Bronze Award

Sandeep Singh

Lead Data Scientist at HP | Personal Systems Quality
Greater Houston

Summary

As a Lead Data Scientist at HP Inc., I have over 8 years of experience in Data Science and Software Engineering, and a Master's degree in Computer Science.
I have a strong proficiency in analytics, and enjoy creating meaningful stories and generating valuable insights from data.
My programming background and expertise in building Machine Learning systems have helped me excel in various data science roles at HP.

Currently, I guide product quality decisions and improvements through telemetry data products, analytical dashboards, and machine learning. With my experience in Data Science and a strong foundation in computer science, I've developed various AI and Machine Learning solutions like preventive maintenance systems, recommendation engines, and sales prediction models. These solutions drive data-informed product design and quality decisions, resulting in substantial efficiency gains and increased revenue.

Prior to my data science career, I worked as a Software Engineer, building production-ready applications using Big Data, Hadoop platforms, and languages such as Java, Python, and SQL.

Key technical skills

Data Analytics: Predictive & Descriptive Modeling, Timeseries Analysis, NLP, Deep learning, Recommendation Engines, Generative AI, LLMs
Cloud Platforms and Tools: Databricks, AWS, Sagemaker, Microsoft Azure, Microsoft PowerBI
Languages: Python (Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn, DeepAR, Langchain, Streamlit), R, Java
Database Technologies: SQL, Hive, Neo4j

If my profile interests you, please feel free to reach out to me here or at sandy91@gmail.com.

Experience

HP

10 years 1 month

Lead Data Scientist, Personal System Quality

June 2024 - Present (2 years)

Greater Houston

Lead Data Scientist, Personal System Quality

June 2023 - June 2024 (1 year 1 month)

Austin, Texas Metropolitan Area

• I lead a team of 5 Data Scientists and Analysts in designing and developing large-scale telemetry data products, including analytical dashboards and machine learning solutions. These products play a pivotal role in guiding product quality decisions and driving improvements.
• I lead the creation of an AI-powered conversational agent capable of interacting with users to deliver information, insights, and support related to telemetry data and its correlation with other dimensions. The agent provides recommendations, data-driven decision support, and troubleshooting with predictive analytics and advanced language models.
• I've mentored and onboarded colleagues, facilitated recruitment efforts, and presented product strategy to architecture review boards and senior leadership. My role blends technical leadership with analytics in shaping data science initiatives within the Personal Systems Quality group.

Senior Data Scientist | Analytics and Data Science Solutions

November 2021 - June 2023 (1 year 8 months)

Austin, Texas, United States

• Analyzed system telemetry data, along with descriptive analytics and ML models for preventive maintenance.
• Built a preventive maintenance machine learning solution in Databricks for predicting battery failure for 300k units, using multiple classification algorithms and system telemetry data.
• Implemented a recommendation engine for optimize part ordering for Personal Systems sales agents. The solution included predictive and classification models and achieved 80% global adoption and 85% part prediction accuracy.

• Also partnered with solution architects to set up a Data Science Platform on Azure with
Databricks and MLFlow, as well as AWS and Sagemaker, onboarding various Data Science projects across HP.
• Throughout my career at HP, I've collaborated with cross-functional teams and vendors, and promoted knowledge sharing by publishing technical articles and conducting learning sessions. This enhanced team expertise and informed leadership decisions.

Data Scientist II | Enterprise Data and Analytics
September 2019 - November 2021 (2 years 3 months)
Austin, Texas, United States

I switched from Software engineering to Data Science to build upon my passion for Data Analytics and Machine learning.
Sales Quota Analytics:
• Enhanced existing prediction systems and models for Sales target setting. Through various
statistical techniques and model fine-tuning, I improved the accuracy of 3 existing ML systems by
10% and also solved some historically challenging prediction problems. Our business
teams also greatly appreciated my work.
• Took the lead to set up a new pipeline for a new part of this Sales Quota Analytics project. This
covered the sales target forecasts for a major region of HP's Sales partners. I conducted the
historical data analysis, model development, historical and production runs, as well as set up the
PowerBI dashboards for data exploration and accuracy analysis. The solution had averaged around
an 85% accuracy rate for quarterly predictions.
• Also tried out Deep learning models(DeepAR and GluonTS), amongst various other ML
techniques to improve the accuracy of my models.

Big Data Software Engineer | Enterprise Data and Analytics
May 2017 - September 2019 (2 years 5 months)
Austin, Texas

After completing my Master's degree, I joined HP Inc. in a full-time role as Big Data/Software Developer.

• I led a team to deliver a Big Data solution to transform and load TB+ of customer, revenue and
other domain data from over 150 Hadoop tables into different downstream systems.
I utilized Talend for developing the ETL components, along with Hive, Java, and SQL.
• I led a project to build the data extracts for one of our newly integrated analytics tools. Collaborating
with various domain teams, I gathered the requirements, analyzed the data as well as
developed and delivered the extracts.
This pipeline consisted of Talend jobs to connect Printer and Toner transactional data between
Oracle and Hive databases.
• Some of my other projects or activities included: implementing Jenkins and Talend to build CI/CD
pipelines for deploying software into production, a code-fix accelerator to update thousands of
Talend jobs without manual effort, various hive and Talend optimization tasks as well as helping
out with a platform upgrade involving over a hundred production jobs.
• I also engaged in various PoCs, workshops, trainings, certifications, bug fixes, etc.

Software Engineer Intern | Core Engineering and Analytics Team

May 2016 - April 2017 (1 year)

Austin, Texas Area

I joined HP Inc. as a graduate intern during my Master's degree.
• As my main internship project, I developed a SQL Reverse Engineering tool coupled with a knowledge
graph in Neo4j of 50k SQL queries.
• For this tool, I built a SQL parser in Java and Antlr to extract column and table level mappings as well
as the relationships between different SQL queries.
• These mappings and all the other content of the SQL queries were fed into Neo4j databases, which
were used to derive data pipeline and query optimization insights.

• I also developed an end-to-end data ingestion and ETL process for HP’s Enterprise Data Lake

pipeline, leveraging tools like Flume, Falcon, Oozie, Pig, and Hive.

• Lastly, I set up Nagios, an open-source remote monitoring tool, to enable real-time monitoring of
HP's printer telemetry servers and their services, and automated alerts to reduce downtime.

Genpact Headstrong Capital Markets
Big Data Engineer- Software Developer
January 2014 - July 2015 (1 year 7 months)
Noida Area, India

I worked in the Big Data R&D team which was a part of the Advanced Technology Center of Excellence(AT-CoE) Cell. As a new grad stepping into the industry for first time, I joined this team to gain experience in an emerging field.

• Built full stack Hadoop based solutions using Map Reduce, Hive and Pig.
• Worked on processes which involved everything from ingesting data to delivering the end solution with a custom user interface.
• Also deployed Hadoop clusters and set up Virtual Machines.

Zapak Digital Entertainment Limited
Software Developer Intern
May 2012 - June 2012 (2 months)

• As an intern at Zapak, an online gaming company, I developed a user interface in Java for employees to view Zapak’s gameplay analytics.
• The interface also allows them to insert, modify, delete gameplay data as well as manage users and admins. Other features included were pagination, export to excel and password encryption.
• I used Java Servlets to develop the backend of application, with JSP(Java Server Pages) for the front end. The app connected to a SQL db to retrieve Zapak's data.

India Meteorological Department
Software Developer Intern
May 2011 - June 2011 (2 months)

During this 6 weeks internship, I developed software in Java to ease the process of removing locks and restarting the DAG(Database Availability Group) servers. It involved provisions for the user to Start, Stop and Check the Status of three DAG servers with a single click.

Education

University of Illinois at Chicago

Master’s Degree, Computer Science · (2015 - 2017)

Amity University

Bachelor of Technology (BTech), Computer Science · (2009 - 2013)

Dayawati Modi Academy-I

· (1998 - 2009)